Preliminary Results from Ionospheric Threat Model Development to Support GBAS Operations in the Brazilian Region

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ABSTRACT

The Brazil ionospheric study project aims to develop a new Ground-based Augmentation System (GBAS) ionospheric threat model which reflects Brazilian low-latitude conditions. This study utilizes dual-frequency GNSS data collected over about 120 active ionosphere days to assess Brazilian ionospheric behavior. Over 1000 anomalous gradients were observed from post-processed data, and almost all of these were caused by night-time Equatorial Plasma Bubbles (EPBs). A significant number of ionospheric gradients exceeded the upper bounds (375 – 425 mm/km) of the Conterminous U.S (CONUS) threat model. In particular, the largest gradient of about 850 mm/km is twice as large as the maximum gradient observed in CONUS. Since a larger bound on maximum gradient would have a significant effect on system performance and availability, more detailed study of the behavior of the most severe ionospheric gradients is needed to establish a low-latitude threat model.

This paper defines a series of threat model parameters to model the geometry of EPBs. The observations from Brazil were examined in detail to quantify both existing parameters used to model ionospheric storms in mid-latitude regions and newly introduced parameters for EPBs. A maximum depletion of 35 meters and transition zone lengths of between 20 and 450 km have been estimated for EPBs which produced the most extreme gradients. These EPBs appeared to be moving roughly eastward and parallel to the geomagnetic equator (within ±35 degrees of magnetic north) with speeds of between 40 and 250 m/s. Almost all significant EPBs occurred during post-sunset hours, between 1800 and 0500 local time, with the peak in the late evening just before local midnight. Defining and modeling these characteristics of EPBs are key to the development of a GBAS ionospheric mitigation and safety case for operational approval in Brazil and other low-latitude locations.

1.0. INTRODUCTION

The Ground-based Augmentation System (GBAS), which enhances the performances of Global Navigation Satellite Systems (GNSS), must be sufficiently robust to ionospheric anomalies and irregularities to avoid hazardous errors and frequent service interruptions. The Honeywell SLS-4000 GBAS installed at Galeão International Airport (GIG) in Brazil was originally configured with the ionospheric threat model for the Conterminous U.S (CONUS), which has a maximum spatial gradient of 425 mm/km [1]. However, Ionospheric activity in equatorial regions (within ±25 degrees latitude of the geomagnetic equator) is known to be significantly more variable and intense than what is encountered (and has been extensively studied) in mid-latitude regions such as CONUS. Models of nominal and threatening ionospheric behavior developed for mid-latitude regions do not generally apply in equatorial regions. This motivated the development of a new threat model for Brazil based upon analyzing data collected in this region.

The Brazil ionospheric study project, initiated in October 2013, was conducted by an international and interagency team comprised of experts on the ionosphere and GBAS from Brazil, the U.S., and elsewhere. The project teams included DECEA, ICEA, INPE, the FAA Technical Center, Stanford University, Boston College, IEAV, UNESP, Mirus, NAVTAC, KAIST, and the Catholic University in Rio de Janeiro. In this project, old and new ionospheric data were collected and processed in order to assess ionospheric behavior covering as much of the Brazilian landmass as possible. Most of the significant ionospheric spatial gradients observed from this data occurred under the presence of Equatorial Plasma Bubbles (EPB) of charged-particle depletion. This provided further motivation to specifically model EPB characteristics so that we can simulate their impact on GBAS users.

Section 2.0 of this paper introduces the dual frequency GPS data sets used for the Brazilian ionospheric study. In Section 3.0, we present the results from the analysis of anomalously large ionospheric spatial gradients within this data. Section 4.0 defines the parameters of a simplified EPB geometry which are proposed for use in GBAS ionospheric impact simulation and shows preliminary results for the possible values of these parameters. Section 5.0 draws conclusions along with recommendations for future work.

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2.0. DATASETS FROM BRAZILIAN GNSS NETWORKS

A comprehensive analysis of Brazilian ionospheric behavior was carried out using dual-frequency code and carrier-phase GNSS measurements (L1 at 1575.42 MHz and L2 at 1227.60 MHz) gathered from Brazilian GPS reference stations. These stations are owned and operated by several organizations, including the Brazilian Network for Continuous GPS Monitoring (RBMC) maintained by the Instituto Brasileiro de Geografia e Estatística (IBGE), the Low-latitude Ionospheric Sensor Network (LISN), the Concept for Ionospheric scintillation mitiGAtion for professional GNSS in Latin America (CIGALA), the Brazilian Airspace Control Institute (ICEA) Septentrio and Trimble receiver network, the Integrated Positioning System for Geodynamic Studies (SIPEG), and Honeywell International (the SLS-4000 receivers in and near GIG). Figure 1 shows the locations of the Brazilian GNSS reference stations used in this study, of which over 180 stations are operational as of 2014.

Different days within these data sets are classified based on their apparent characteristics and severity of ionospheric behavior. About 120 days during the three-year period from April 2011 to March 2014 (which

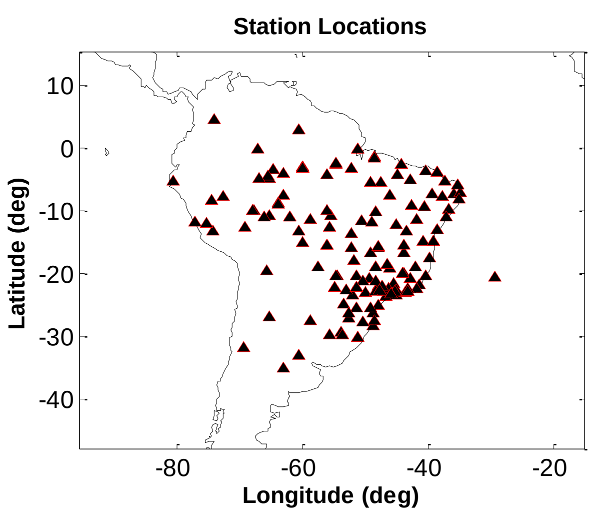

Figure 1. Locations of the Brazilian GNSS reference stations used in this study (as of 2014).

includes the peak of the current solar cycle) were selected for detailed study. This was done by utilizing pre-existing measures of space weather intensity, including planetary K (Kp) and disturbance/storm time (Dst) values. While these geomagnetic indices characterize global ionospheric disturbances fairly well, they are not the best measures of local ionospheric distortions. Thus, we particularly focus on the L1-band amplitude scintillation parameter, S4, to identify post-sunset equatorial ionospheric instabilities such as EPBs. The selected dates for detailed ionospheric study include 85 scintillating days (based on the depth of amplitude scintillation and number of satellites affected by scintillation during the same day at different locations within Brazil) and 30 storm days (based on Kp and Dst indices). Eight (8) non-scintillating days were also selected to be used as a reference when testing the post-processing software described below that estimates ionospheric delays and gradients.

3.0. ANOMALOUSLY LARGE IONOSPHERIC SPATIAL GRADIENTS

GPS dual-frequency observations over 24-hour periods from the selected data sets were input into the Long-Term Ionospheric Anomaly Monitor (LTIAM) software package. This tool was designed to continually monitor anomalous ionospheric behaviors over the life cycle of GBAS [2]. It performs automated data downloading, formatting, measurement-error screening, ionospheric delay and spatial gradient estimation, and anomalous gradient search and analysis.

Preliminary results from the LTIAM processing using the Brazilian GPS observation data generated 1135 threat points that exceeded a threshold (slant) gradient of 200 mm/km in the initial gradient estimates. Note that the final gradient estimates could be lower than 200 mm/km after manual validation. These 1135 threat points are represented along with their corresponding satellite elevation angles in Figure 2. The gradients estimated from the Brazilian GNSS data and the upper bound of the CONUS threat model (for comparison) are denoted with blue crosses and a dashed line, respectively. The maximum ionospheric gradient observed in this study was 850.7 mm/km (this and all gradients reported below are in the slant domain). This magnitude is more than twice that of the largest gradient of 413 mm/km observed in CONUS [1].

Including the largest gradient, 31 separate events had ionospheric spatial gradients exceeding 500 mm/km. Because each event was checked for internal consistency with ionospheric behavior (to exclude the possibility of receiver measurement errors), the large number of gradients exceeding 500 mm/km demonstrates that ionospheric anomalies creating gradients larger than those found in CONUS can definitely occur in Brazil. The five

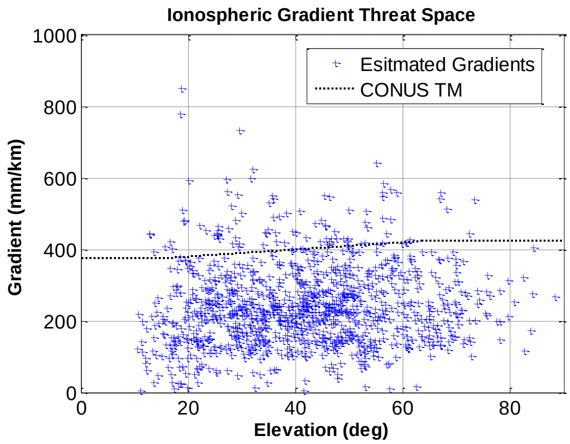

Figure 2. Brazilian ionospheric gradient threat space. The estimated gradients from the Brazilian GNSS data are denoted with blue crosses, while the upper bound of the CONUS threat model [1] is marked with a dashed line.

largest gradients, ranging from 600 to 850 mm/km, were further validated by the method proposed in [3] to be caused by Equatorial Plasma Bubbles (EPBs) of charged-particle depletion that induced sharp ionospheric vertical delay drops of about 8 – 15 meters in L1 signals.

Relative to the CONUS threat model, higher ionospheric gradients were observed at lower elevations in the Brazilian region. However, this elevation dependence is based on limited gradient data. The spatial characteristics of ionospheric disturbances should be further studied to better determine satellite geometry dependency, including satellite azimuth as well as elevation angles.

Evening pre-reversal-enhancement associated Raleigh-Taylor instability causes the formation of EPBs, resulting in steep depletions in ionospheric Total Electron Content (TEC) after sunset [4][5]. The histogram in Figure 3 shows the occurrence times of large gradients observed in

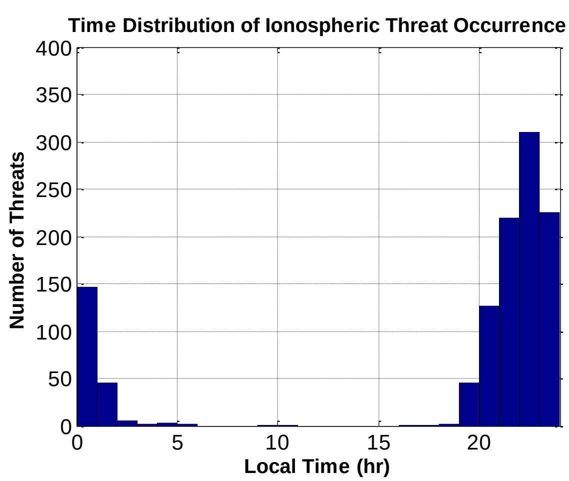

Figure 3. Occurrence times of large ionospheric spatial gradients.

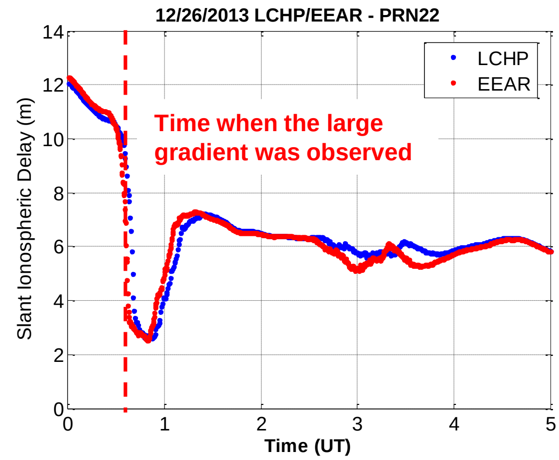

Figure 4. An example of EPB-induced ionospheric depletion on L1 signals: Dual-frequency estimates of slant ionospheric delays observed from stations LCHP and EEAR viewing PRN 22 on 26 December 2013.

the Brazilian region. Almost all of the significant gradients occurred during the post-sunset hours, between 18:00 and 02:00 (Local Time; LT), with the peak in between 22:00 and 23:00 (LT). Furthermore, these gradients were observed mostly during steep fall-offs (or depletions) of ionospheric delay, as shown in Figure 4, or corresponding recoveries in delay within or on the trailing end of EPBs. Thus, it is very likely that the occurrence of severe ionospheric spatial gradients in the Brazilian region is associated with night-time EPBs.

Figure 5 shows a vertical ionospheric delay map at 01:04:00 UT on 01 March 2014 in the region around São José dos Campos, Brazil. Slant ionospheric delays measured from Brazilian stations viewing multiple satellites are converted to equivalent vertical delays

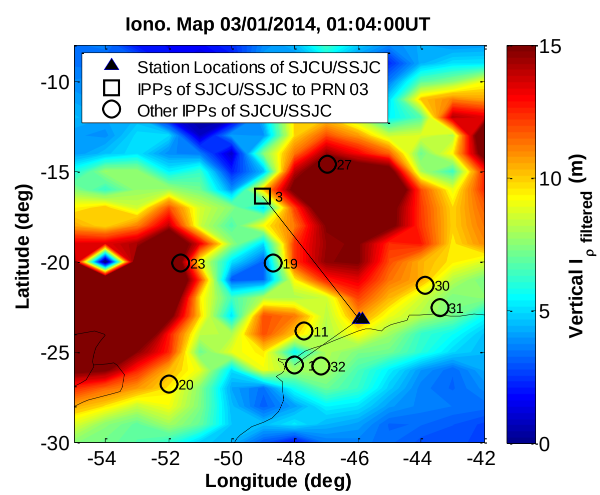

Figure 5. Two-dimensional ionospheric delay map at 01:04:00 UT on 01 March 2014, when the largest gradient of 850.7 mm/km was observed from two stations SJCU and SSJC (triangles) viewing PRN 03 (square).

assuming a hypothetical ionosphere thin shell model with a height of 350 km. A color contour map is created by interpolating scattered Ionospheric Pierce Points (IPPs) with their vertical delays over a 1-by-1 degree rectangular grid. An IPP is defined as a point at which the line-of-sight between satellite and receiver intersects the thin-shell model of the ionosphere. A vertical delay depletion of 14 meters was observed from two stations SJCU and SSCJ viewing PRN 03, as an EPB moved roughly eastward around 01:00 UT. The IPPs of PRN 03 (denoted with squares) from these two stations encountered the EPB transition zone in sequence, experiencing delay depletions of about 35 meters in the slant domain. The resulting slant gradient between the two stations (which are about 10 km apart) was 850.7 mm/km at 01:04:00 UT. This is the highest gradient observed in the Brazilian GNSS data examined from April 2011 to March 2014.

The IPPs of other satellites seen from these two stations were also under the influence of the same EPB, as shown in Figure 5. If it is possible for multiple satellites to be affected by large gradients from the same EPB at the same time, this needs to be included in the EPB threat model. The typical time between the final approach fix and the minimum CAT-I decision height at 200 ft is about 150 seconds [6]. Thus, we investigated simultaneous satellite impacts of a single EPB event within a time window of ± 150 seconds of the time when the largest gradient was observed. The results summarized in Table 1 show that two or more satellite impacts are possible during a single CAT-I precision approach. The IPPs of PRN 19 passed through the same EPB transition zone as those of PRN 03, observing an extreme gradient of 571 mm/km about 150 seconds prior to 01:04:00 UT. In contrast with PRN 03 and PRN 19, the IPPs of PRN 11 were affected by an ionospheric irregularity inside the EPB depletion region. The resulting gradient was 465.0 mm/km, which occurred one minute after PRN 03 observed the gradient of 850.7 mm/km.

Table 1. Maximum ionospheric gradients observed from SJCU and SSCJ viewing multiple satellites within ± 150 seconds of 01:04:00 UT.

PRN	Time (UT)	Gradient (mm/km)
19	01:01:30	571.0
3	01:04:00	850.7
11	01:05:00	465.0

4.0. ANALYSIS PROCEDURE FOR EPB THREAT MODEL PARAMETERS

The results shown in the previous section imply that the upper bound on spatial gradients for the Brazilian ionospheric threat model needs to be considerably higher than that of the CONUS model. A larger gradient bound applied to the same geometry as the CONUS model would have a significant effect on system performance and availability. However, previous knowledge of ionospheric physics [4][5][7][8] plus observations from Brazil like those shown in Figures 4 and 5 demonstrate that EPB characteristics are much different than those of ionospheric storms in CONUS. Hence, a more detailed study of the behavior of the most severe ionospheric gradients in Brazil is needed to create a representative threat model, and different parameters need to be defined to represent behavior that is specific to EPBs.

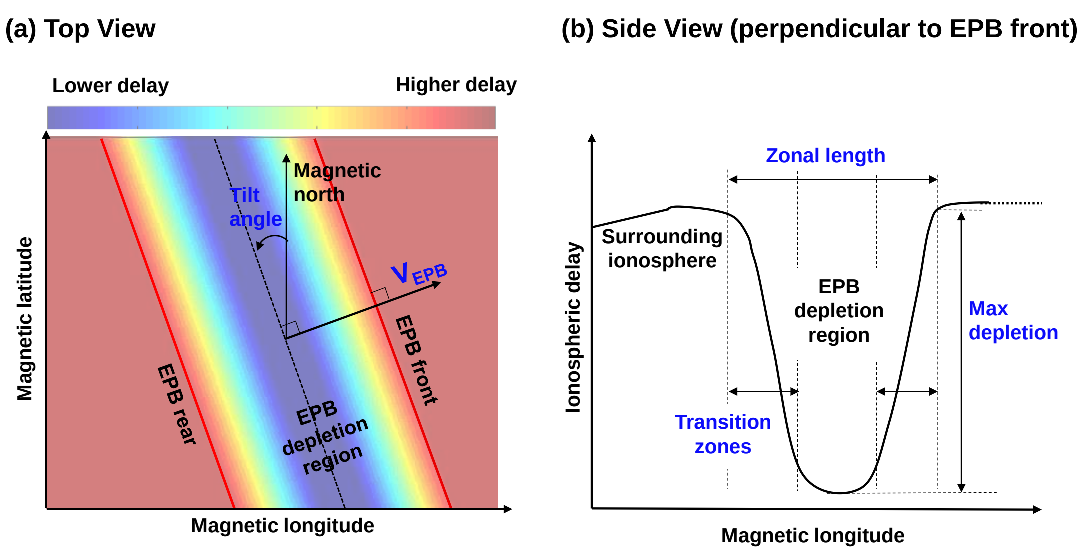

Figure 6. Definitions of EPB threat model parameters, including tilt angle, propagation speed, zonal length, transition zone, and max depletion.

Table 2. Summary of EPB parameter estimation results (for gradients > 400 mm/km)

Parameter	Velocity (m/s)	Tilt Angle (deg) ※ measured from geomagnetic north	Transition Zone Length (km)		Zonal Length (km)	Max Depletion (m in slant domain)	
			Primary	Total		Primary	Total
Minimum Value	39.6	-34.9	11.25	22.1	95.9	3.0	1.6
Maximum Value	246.2	33.7	185.5	454.2	751.7	19.6	35.2

In this section, we first investigate the characteristics of EPB movement and spatial structure and then define a simplified EPB model and parameters to represent this behavior. EPBs that occur after sunset are known to move eastward [7]. While being aligned with geomagnetic field lines, EPBs have a band-like structure elongated over a few thousand kilometers with a relatively smaller zonal length, or distance across the depletion region, of a few hundred kilometers [8]. EPBs are typically oriented perpendicular to the geomagnetic equator and move parallel to it. These well-known characteristics of EPBs can be approximated by defining parameters of their movement and structure. The EPB model shown in Figure 6 has a spatially linear wave front and rear in a meridian direction with a constant eastward propagation speed. Figure 6 illustrates this model. To describe the EPB orientation and motion, two parameters are defined in Figure 6(a): (1) tilt angle relative to the geomagnetic equator, (2) speed with respect to the ground. To model the EPB structure, we define three more parameters in Figure 6(b): (3) gradient transition zone length, (4) zonal length, (5) maximum depletion (in the slant domain).

The velocity and tilt angle of EPBs are computed using a three-station-based trigonometric method (see [9] for details). This algorithm assumes that a semi-infinite ionospheric front with a spatially linear change in delay moves in a constant speed with no change in direction. However, this assumption does not generally hold for EPBs. Furthermore, the method requires more than three stations which are closely located with good observation geometry. Thus, we introduced a second method for velocity estimation which utilizes a regional ionospheric delay map of the form shown in Figure 5. The eastward speed of EPB motion is estimated by comparing specific points on curves of ionospheric delay different times. These estimated speeds are compared with the results from the three-station-based method to observe the degree of difference, which turns out to be relatively small.

Table 3. Comparison of the (preliminary) Brazilian threat model with the CONUS threat model

Threat model parameter (or characteristics)	CONUS threat model	Brazilian threat model
Anomaly time of day	Any time	Local night-time
Anomaly approach direction	Any direction	Within title angle of magnetic equator (W-E)
Multiple-satellite impacts	Worst pair of 2 SVs	Multiple SVs depending on EPB and IPP locations
Maximum ionospheric gradient	425 mm/km	860 mm/km
Speed	0 – 750 m/s	40 – 246 m/s
Maximum depletion (or max delay)	50 m	20 m (primary) 35 m (total)
Transition zone length (or width)	25 – 200 km	11 – 186 km (primary) 22 – 454 km (total)

Regional ionospheric delay maps and precise estimates of slant ionospheric delay were also used to measure the parameters of EPB zonal structures. We first investigated the longitudinal profiles of ionospheric delay at the latitudes of IPPs where extreme gradients were observed. Then we measured two different longitudes at which the EPB-induced delay depletions start and end. The horizontal distance between them is calculated, and this distance is projected onto a direction perpendicular to the EPB front using the tilt angle estimated from the three-station method to obtain the zonal length of the EPB.

In the estimation of transition zone length, we define two sub-parameters: the total transition zone and the primary transition zone. The total transition zone is a region located between the surrounding ionosphere (the maximum delay) and the minimum TEC zone (the minimum delay) of the EPB, or vice versa. The primary transition zone is a subset of the total transition zone where the largest gradient was observed and the shape of the gradient remains consistent. Some gradient events are consistent from top to bottom; thus the total and primary transition zones for them are the same. We estimate transition zone lengths in the same manner as the larger zonal lengths.

Once transition zone are defined for a given event, we measure the magnitudes of total and primary delay depletions. These estimations are performed using dual-frequency carrier-derived delay estimates. During large delay drops or recoveries, data outages or cycle slips frequently corrupt carrier-phase observations, which calls into question the reliability of the delay estimates. Thus, we use moving-average-filtered code estimates of slant ionospheric delays to validate carrier-derived estimates by comparing the trends of the two estimates.

5.0. RESULTS FROM THREAT MODEL PARAMETER ANALYSIS

The preliminary results of EPB parameter estimation are summarized in Table 2. These results include all threat events whose validated (slant) gradients are above 400 mm/km. The EPB fronts which generated such severe gradients moved eastward with speeds of 40 to 246 m/s with respect to the ground and were oriented with tilt angles of ±35 degrees measured from geomagnetic north. The zonal scale of EPB events varies from 96 km to 750 km. The transition zone lengths were between 11 and 186 km for the primary transition zone and between 22 and 454 km for the total transition zone. Both primary and total depletions are less than 35 meters in the slant domain.

To develop a viable threat model which captures ionospheric anomalies in the Brazilian region, new parameters and constraints are introduced in addition to the parameters used for the CONUS model. The upper bound (425 mm/km) on the gradient threat space for CONUS (and mid-latitudes in general) is insufficient to covering EPB-induced threats in equatorial regions. This can cause loss of system availability for CAT-I precision approaches at low-latitudes. However, as noted above, the behavior of EPBs and large gradients induced by them is qualitatively different than that of the ionospheric storms observed to generate the largest gradients in CONUS. For example, the total transition zone length of EPBs varies considerably compared to the front width of storm-driven gradients in the CONUS threat model, and the total spatial extent of EPBs (zonal length) is smaller than regions affected by ionospheric storms. Since multiple satellites can be affected by the same EPB, and multiple EPBs can exist in the same region of space, more-than-two-satellite impact cases may need to be considered for the Brazilian threat model, as opposed to the worst two-satellite impacts that are simulated for GBAS in CONUS (see [10]).

Among the EPB parameters analyzed above, occurrence time, tilt angle, propagation speed (eastward), and maximum depletion are more constraining compared to the corresponding parameters of the CONUS threat model (see the comparison of Brazil and CONUS parameters in Table 3). The approach direction of EPBs is from west to east within 35 degrees of the geomagnetic equator. The range of speeds is smaller than that of ionospheric-storm-induced anomalies in CONUS. The maximum depletions (primary and total) in the slant domain are smaller than the 50 meters assumed for maximum differential delay in the CONUS model). These EPB parameters should be incorporated into the threat model and the worst-case ionospheric anomaly impact simulation (explained in [10] [11]).

6.0. CONCLUSIONS

Anomalous ionospheric behavior over the Brazilian region has been studied for the peak of the current Solar Cycle #24. The maximum ionospheric spatial gradients discovered in Brazil are considerably higher than those in CONUS. This will likely have a significant effect on GBAS availability. Analysis of other parameters forming a geometric model of equatorial plasma bubble (EPB) impacts has been completed. The estimated parameters, including spatial gradients, form a preliminary Brazilian ionospheric threat model that will be incorporated into GBAS impact simulations for the Brazilian region. Depending on the results from these impact studies, it may be necessary to develop additional ionospheric mitigation strategies that reduce the potential loss of GBAS availability in the Brazilian region.

However, the preliminary results were derived only from the threat events observed to date for which ionospheric gradients were larger than 400 mm/km. Thus, further examination of data from Brazil will be conducted along with integrated analysis by ionospheric and GBAS experts. In addition, the definition and analysis of additional geometric parameters for EPBs would help improve the current Brazilian threat model. Two key parameters to be added are the maximum longitudinal extent of EPB gradients (i.e., along the axis near magnetic north in Figure 6(a)) and the minimum distance between successive separate EPBs which may simultaneously affect the same GNSS satellite geometry.

Finally, while EPBs are known to create large spatial gradients throughout the equatorial regions of the Earth, it is not yet known whether maximum gradients elsewhere in these regions are as large as those observed in Brazil. Studies similar to this one performed in Brazil should be conducted for other low-latitude regions that plan to employ GBAS or other single-frequency GNSS augmentation systems.

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